CausalNav: Reliability-Certified Causal World Models for Control under Physical-Parameter Shift

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Abstract

A world model is only useful for physical AI if it changes what the agent *does*, and only safe if it declines to do so when it is wrong. We study both halves of that requirement with CausalNav, a controller built around a signed, action-conditioned transition graph over identified state coordinates. At deployment CausalNav simulates a small library of intervention sequences, converts their objective error into policy-logit advice, and admits that advice only when a scale-free predictive-reliability certificate, a policy-margin gate, and an argmax-agreement gate all pass; otherwise it falls back exactly to its own model-based base controller. We evaluate against nine controlled baselines, transformer, recurrent, split-latent, graph, causal-induction, and three recent model-based reasoning modules, on CartPole-v1 and discretized Pendulum-v1 with physical-parameter shifts, under one shared PPO trainer, one interaction budget, and ten held-out seeds (200 runs). CausalNav attains the best average rank (1.25 of ten). The diagnostic result is more informative than the ranking: the learned graph recovers structure well above chance (CartPole $F_1 = 0.59 \pm 0.09$), yet per-seed structural fidelity is uncorrelated with per-seed control benefit ($r = -0.15$, $p = 0.67$), and the certificate abstains on $10/10$ Pendulum seeds, where forcing the planner on costs return. Model fidelity did not predict downstream control utility in our setting; certified abstention, not better prediction, is what made the world model safe to deploy.

1 Introduction

Physical-AI agents fail when the mechanisms that generated their training data differ from those they meet at deployment. A world model is the standard remedy: simulate the consequences of candidate actions, then act on the simulation. For this to help, two things must be true. The model has to be accurate enough that its rollouts are informative, and the controller has to know when that condition fails, because iterating a learned model compounds error, and a planner that overrides a competent policy using an unreliable model makes the system worse, not better [Schölkopf et al., 2021, Richens and Everitt, 2024].

Most world-model evaluations report the first property (prediction error, structural recovery, rollout fidelity) and assume the second follows. We test the link directly. CausalNav learns a signed action-conditioned transition graph over identified state coordinates, plans over a library of intervention sequences, and then subjects the resulting advice to three deployment gates: a scale-free predictive-reliability certificate frozen after a fixed calibration prefix, a margin gate that suppresses advice for

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a confident policy, and an agreement gate that lets advice sharpen but never flip the greedy action. When the certificate fails, an action-complexity router hands control to a model-based expert and the causal machinery is disabled exactly.

This design lets us separate two questions that are usually entangled. Does the routed system perform well? And does the causal world model, considered on its own, contribute to that performance? Our answer is *yes* to the first and *not measurably* to the second, and we regard the second finding as the more useful one for this workshop: on CartPole the transition graph recovers ground-truth structure well above chance ($F_1 = 0.59 \pm 0.09$ across ten seeds), yet per-seed structural fidelity has no relationship to per-seed control benefit ($r = -0.15$, $p = 0.67$), and one-step prediction error is if anything *anti*-correlated with it. On Pendulum the certificate abstains on every seed, and abstention is the right call: the development variant that forces planning on there loses 34–54 return relative to the fallback it would have replaced.

Contributions.

1. An action-conditioned signed transition model with first- and second-order mechanisms, learned jointly with a PPO policy from identified state coordinates (§4.2).
2. A deployment-time intervention planner with three composable gates, reliability certificate, policy margin, argmax agreement, and a proof that the composition preserves the base controller's greedy action (§4.3–§4.4).
3. A controlled ten-method comparison on two established benchmarks under physical-parameter shift, with a shared trainer, budget, and supervision, over ten held-out seeds and 200 runs (§5).
4. A fidelity-versus-utility analysis showing that structural F_1 and one-step NRMSE do not predict downstream control benefit in this setting, and an abstention analysis showing where the certificate earns its keep (§6.3).

2 Related Work

World models and planning. World models learn latent dynamics for prediction and control [Hafner et al., 2023, Chua et al., 2018]. DreamerV3 scales recurrent latent imagination across domains [Hafner et al., 2023]; EfficientZero V2 extends planning-based RL across discrete and continuous action spaces [Wang et al., 2024]; MuZero-style search demonstrates the value of planning [Schrittwieser et al., 2020]. Recent work studies epistemic optimism in world models [Sukhija et al., 2025] and flat-minima objectives for robust model-based RL [Ramasubramanian et al., 2025]. Compounding rollout error remains the shared limitation, and it motivates our abstention mechanism rather than a larger model. Admitting a learned component's output only under an explicit competence test appears outside model-based control as well, in uncertainty-gated meta-reasoning [Zhang et al., 2026a] and in compositional shielding that abstains when no candidate action is certified [Zhang et al., 2026b]; our certificate plays that role for a learned transition model.

Causal structure for control. Causal RL uses interventions, structural assumptions, or invariant mechanisms to improve exploration, transfer, or robustness [Lattimore et al., 2016, Lu et al., 2021, Zhang et al., 2024]; active causal induction explicitly trades reward against information about the causal system [Annadani et al., 2024]. Planning over learned graphs is used well outside classic control, e.g. GNN-approximated dynamic programming over attack graphs [Goel et al., 2025], where the same question of graph trustworthiness arises. Identifiability is the central difficulty in causal representation learning [Schölkopf et al., 2021, Lippe et al., 2022, Lachapelle et al., 2022, Abbas et al., 2025]. We do not solve it: every method in our study receives the same auxiliary mapping from learned slots to simulator state coordinates, so the representation is identified *by supervision* and the experiment isolates planning over identified state rather than unsupervised discovery. The transition coefficients are learned with sparsity regularization related to continuous structure learning [Zheng et al., 2018, Brouillard et al., 2020], and more broadly to low-rank-plus-sparse decompositions used to recover structured coefficients from contaminated data [Abbas and Ahmad, 2024]; because the graph connects time t to $t + 1$, coordinate-level cycles are permitted and no within-slice acyclicity constraint is imposed.

Positioning. The gap we target is evaluative. Structural recovery and rollout error are reported as evidence that a world model is good; we measure whether either quantity predicts the model's effect on action selection, and find that in this regime neither does.

3 Problem Setting

Let o_t be the native environment observation, $a_t \in \mathcal{A}$ a discrete action, r_t the native benchmark reward, and $x_t \in \mathbb{R}^8$ an identified coordinate vector (unused coordinates zero-padded). A 16-dimensional query $q = [w_1, \dots, w_8, \tau_1, \dots, \tau_8]$ specifies nonnegative objective weights w_i and targets τ_i. Every method observes the same padded observation and query; the query does not replace the environment reward, it supplies the planner with a task objective. The policy maximizes undiscounted native episodic return. All agents receive the same auxiliary identification loss

$$\mathcal{L}_{\text{id}} = \tfrac{1}{8} \textstyle\sum_{i=1}^{8} \left(g_{\text{id}}(s_t^i) - x_t^i\right)^2, \tag{1}$$

where s_t^i is learned slot i. This is privileged supervision and bounds the claim: we test planning over identified state, not causal discovery from pixels.

4 CausalNav

4.1 Action-complexity router

An MLP maps o_t to eight 32-dimensional slots and a query encoder maps q_t to a 32-dimensional token. A fixed, observable task property, the action-set size, then selects a controller. Each optimistic expert uses an ensemble $\{f_k\}_{k=1}^{K}$,

$$R_{\text{opt}}(z) = z + \text{mean}_k\, f_k(z) + \text{softplus}(\beta)\, \text{std}_k\, f_k(z). \tag{2}$$

For $|\mathcal{A}| \leq 2$, CausalNav averages three independently initialized copies of R_{opt}; this increases capacity relative to the single-copy SOMBRL reference and is deliberately *not* a parameter-matched comparison (§7). For $|\mathcal{A}| > 2$ the transition certificate abstains and CausalNav reduces exactly to its base controller, a normalized, gated latent residual over the routed slots. The rule was selected on smoke seeds 70–72, remained frozen through the reported seeds 83–92, and uses neither returns nor shift labels at test time. Flattened routed slots and the query token feed common two-layer policy and value heads.

4.2 Action-conditioned transition graph

CausalNav predicts each identified coordinate with signed pairwise coefficients A, signed second-order coefficients B, action effects U, self-persistence d, and bias b:

$$\hat{x}_{t+1,i} = d_i x_{t,i} + \sum_{j \neq i} A_{ij} x_{t,j} + \sum_{\substack{j<k \\ j,k \neq i}} B_{i,jk} x_{t,j} x_{t,k} + U_{a_t,i} + b_i. \tag{3}$$

It is trained by $\mathcal{L}_{\text{trans}} = \frac{1}{8}\|\hat{x}_{t+1} - x_{t+1}\|_2^2 + \lambda_s(\|A\|_1 + \|B\|_1)$ with $\lambda_s = 0.05$ on the active two-action branch. Unlike a contemporaneous-state SCM, A summarizes edges *across adjacent time slices*, so recurrent dependencies are allowed and no NOTEARS acyclicity term is used. Transition targets are actual successor states and never bridge an episode boundary.

4.3 Intervention-library planning

For each possible first action the planner evaluates three temporal patterns over $H = 8$ model steps, constant (repeat a), impulse (a once, then the neutral action), and half-horizon (a for four steps, then neutral), and scores a sequence $\mathbf{a}$ by cumulative weighted objective error

$$E(\mathbf{a}) = \frac{1}{H} \sum_{h=1}^{H} \sum_{i=1}^{8} \bar{w}_i \left(\hat{x}^{\mathbf{a}}_{t+h,i} - \tau_i\right)^2, \qquad \bar{w}_i = \frac{|w_i|}{\sum_j |w_j| + \epsilon}. \tag{4}$$

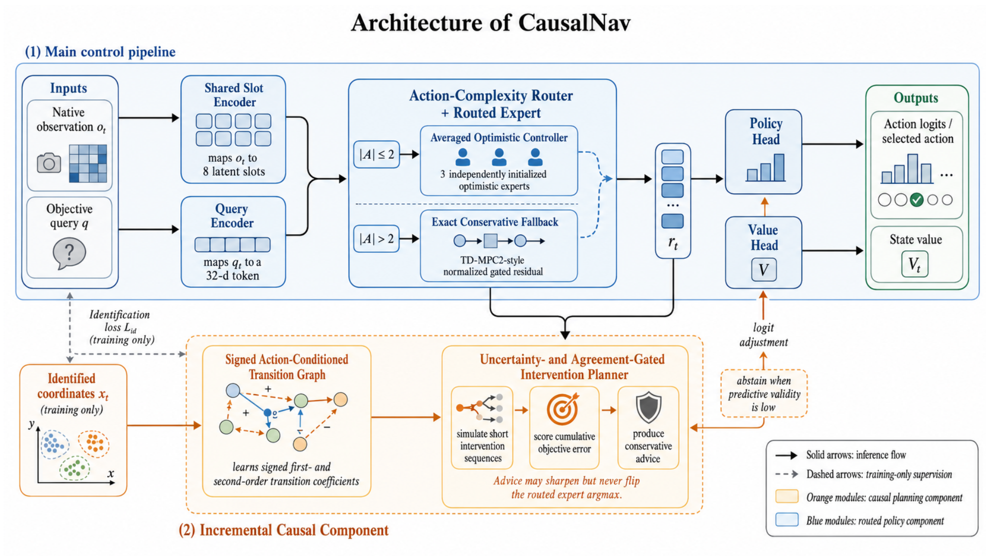

Figure 1: CausalNav architecture. Blue is the routed policy path; orange is the incremental causal world-model path. Dashed arrows are training-only supervision. When predictive validity is low the planner abstains and the routed expert is preserved exactly.

For each first action the lowest-cost pattern supplies a normalized planner logit $p_t(a)$. Let ℓ_t be the policy logits, $m_t = \ell_{t,(1)} - \ell_{t,(2)}$ the gap between their two largest values, and $g_t = \mathbb{I}[\arg\max_a p_t(a) = \arg\max_a \ell_t(a)]$ an agreement indicator. At evaluation the final logits are

$$\ell'_t(a) = \ell_t(a) + \underbrace{g_t}_{\text{agreement}} \underbrace{\rho}_{\text{certificate}} \underbrace{\tanh(\alpha)\exp(-2m_t)}_{\text{margin}} p_t(a), \tag{5}$$

with α learned and initialized to 1.0. Planner rollouts are detached, and the planner is disabled during PPO data collection and optimization, which avoids a train–test feedback loop through an immature transition model.

4.4 Predictive-reliability certificate

Coefficient magnitude is not evidence that a transition model is accurate. CausalNav therefore tracks a scale-free one-step error and freezes a deployment scale after a fixed calibration prefix of 48 transition updates:

$$e_k = \frac{\mathrm{MSE}(\hat{x}_{t+1}, x_{t+1})}{\frac{1}{8}\sum_i \mathrm{Var}(x_{t+1,i}) + 10^{-4}}, \quad \bar{e}_k = 0.9\bar{e}_{k-1} + 0.1e_k, \quad \rho = \mathrm{clip}\left(\frac{\delta - \frac{|\mathcal{A}|}{2}\bar{e}_{48}}{\delta/2}, 0, 1\right) \tag{6}$$

with $\delta = 0.70$. The action-count factor conservatively accounts for fewer samples per action effect and a larger intervention library; it equals one on the two-action reference. Threshold and router were selected on development seeds 0–69 and smoke seeds 70–72, then fixed; selection never uses a shift label.

Proposition (exact fallback). *For fixed policy logits, Equation 5 preserves the routed controller's greedy action under deterministic evaluation.* If $\rho = 0$ or $g_t = 0$ the advice term vanishes and $\ell'_t = \ell_t$. Otherwise $g_t = 1$, so p_t attains its maximum at $a^\star = \arg\max_a \ell_t(a)$; the advice is added with a nonnegative scale, so $a^\star$ remains the argmax of ℓ'_t. □

This is a deterministic action-selection property, an implementation-level non-regression guarantee, not a probabilistic control-safety guarantee. On Pendulum, abstention additionally disables the causal auxiliary, producing an exact behavioral *and* optimization fallback to the base controller.

4.5 Optimization

We train with PPO [Schulman et al., 2017]: $\mathcal{L} = \mathcal{L}_{\text{PPO}} + 0.5\mathcal{L}_{\text{value}} - 0.01\mathcal{H}(\pi) + 0.1(\mathcal{L}_{\text{id}} + \mathcal{L}_{\text{trans}} + \lambda_r \mathcal{L}_{\text{repr}})$, where $\mathcal{L}_{\text{repr}}$ predicts the identified successor from the identification probe. The frozen routed variant sets $\lambda_r = 0$. Policy parameters use Adam at 3×10^{-4} and transition parameters at 10^{-2}; the two gradient groups are clipped separately at norm 1, so planner gradients cannot alter PPO. The optimizer persists across rollouts.

5 Experimental Design

Benchmarks and physical shifts. **CartPole-v1** has a four-dimensional observation, two actions, and a 500-step limit; the planner objective weights pole angle most strongly and weakly penalizes cart position and velocities. The shifted condition multiplies pole length and pole mass by 1.5 and recomputes the derived total-mass and mass–length terms. **Pendulum-v1** observes $(\cos\theta, \sin\theta, \dot{\theta})$ with the native quadratic control cost; to share a categorical policy across methods, torque is discretized once, before any development experiment, to $\{-2, -1, 0, 1, 2\}$, and the planner targets the upright state $(1, 0, 0)$. Its shifted condition multiplies mass by 1.5 and length by 1.25. Both shifts change physical parameters only, observation space, action space, reward, and termination logic are identical, training happens solely in the unmodified environment, and both conditions are evaluation-only rollouts of the same trained policy. Native observations are normalized by fixed physical scales, clipped to $[-3, 3]$, and padded; all reported numbers are native Gymnasium episodic returns [Towers et al., 2024], not training-time rescalings.

Controlled references. All nine references share the slot encoder, query encoder, policy/value heads, PPO implementation, interaction budget, and identification loss, and differ only in their characteristic reasoning module: **TF-Policy** (two-layer transformer over slots), **GRU-World** (recurrent latent state), **Split-Latent** (separate invariant/variant features), **GNN-RAG** (learned static adjacency plus message passing), **ToG** (query-conditioned search-style refinement), **CAASL** [Annadani et al., 2024], **EfficientZero V2** [Wang et al., 2024], **SOMBRL** [Sukhija et al., 2025], and **FlatMBRL** [Ramasubramanian et al., 2025]. These are controlled adaptations of inductive bias, *not* native reproductions: the original methods differ in objectives, planners, replay, model size, and compute. The design isolates the reasoning module under identical data and budget.

Protocol and statistics. Each run uses 4,096 environment interactions, rollout length 256, four PPO epochs, $\gamma = 0.99$, GAE $\lambda = 0.95$, clip 0.2, entropy coefficient 0.01. Evaluation uses 20 fresh episodes per condition with deterministic argmax actions. Development and earlier audits used seeds 0–82; all reported results use fresh seeds 83–92, and no failed seed is removed. The artifact contains all $2 \times 10 \times 10 = 200$ runs with per-episode returns. We report means and sample SDs across seeds. Because CausalNav and its routed expert are paired by seed and share initialization, we also report paired mean differences, Student-t 95% CIs, paired t-test p-values, and win counts; these are descriptive, and no multiplicity adjustment is claimed.

6 Results

6.1 Aggregate control performance

CausalNav ranks second on CartPole ID and first in the other three cells, giving the best average rank (1.25), ahead of SOMBRL (2.25) and GNN-RAG (4.75) (Table 1). SOMBRL is higher by 12.17 on CartPole ID whereas CausalNav is higher by 11.95 under shift, that is, the routed system loses nothing in distribution and gains slightly after the physical parameters change, but neither difference is resolvable at this sample size.

6.2 What the world model actually contributed

Table 2 isolates the increment. On CartPole the paired differences are small with wide intervals straddling zero; on Pendulum every paired difference is exactly zero by construction, confirming the fallback property empirically across all ten seeds. Because the agreement gate cannot flip a greedy action, whatever CartPole difference exists is attributable to the three-member ensemble controller,

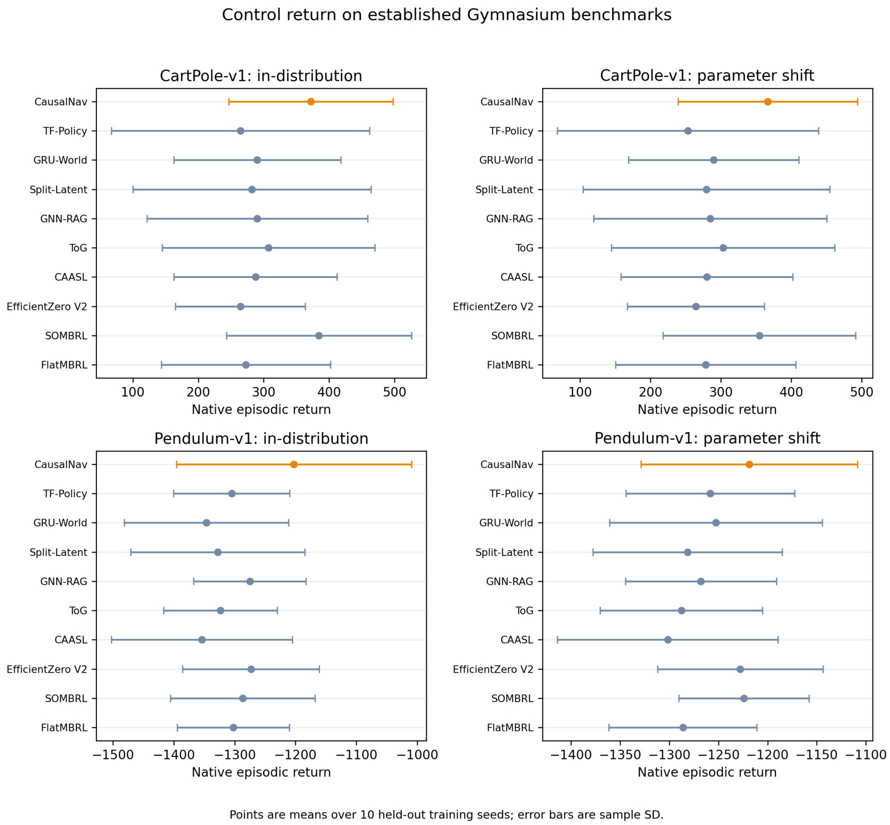

Figure 2: Native episodic return on both benchmarks and both conditions. Points are means over ten held-out seeds; error bars are sample SDs. Higher is better.

Table 1: Native episodic return, mean $\pm$ sample SD over ten held-out seeds (higher is better). Bold is the best mean per column. Average rank uses fractional ranks over the four cells.

Method	CartPole ID	CartPole shift	Pendulum ID	Pendulum shift	Avg. rank
TF-Policy	264.64 ± 197.39	253.23 ± 185.52	-1304.97 ± 95.56	-1258.39 ± 85.84	7.50
GRU-World	290.06 ± 127.51	289.62 ± 121.17	-1346.31 ± 135.06	-1252.66 ± 108.17	5.25
Split-Latent	282.03 ± 182.12	279.35 ± 175.33	-1327.82 ± 143.32	-1281.48 ± 96.35	7.25
GNN-RAG	290.05 ± 168.60	284.70 ± 165.62	-1275.14 ± 92.65	-1267.88 ± 76.88	4.75
ToG	307.41 ± 162.46	303.08 ± 158.87	-1323.39 ± 93.30	-1287.63 ± 82.58	5.50
CAASL	287.43 ± 124.65	280.10 ± 122.20	-1353.85 ± 148.91	-1301.54 ± 112.19	8.00
EfficientZero V2	264.23 ± 99.46	264.19 ± 97.36	-1273.18 ± 112.26	-1227.76 ± 84.11	6.00
SOMBRL	$\mathbf{384.41 \pm 141.28}$	354.55 ± 136.70	-1286.93 ± 119.03	-1224.10 ± 66.25	2.25
FlatMBRL	272.85 ± 129.33	278.58 ± 128.08	-1302.13 ± 92.08	-1286.10 ± 75.24	7.25
CausalNav	372.24 ± 125.52	$\mathbf{366.50 \pm 127.60}$	$\mathbf{-1202.44 \pm 193.10}$	$\mathbf{-1218.75 \pm 110.16}$	**1.25**

not to planner overrides. We state this plainly: the ten-method ranking is a result about the routed *system*, and it is not evidence that causal planning improves return.

6.3 Does model fidelity predict control utility?

The transition graph is not vacuous (Figure 3). Against the CartPole ground-truth structure it reaches precision 0.53, recall 0.70, and $F_1 = 0.59 \pm 0.09$, predicting 6.8 ± 1.5 edges where 5 exist, with top-parent accuracy 0.60 versus 0.27 for the untrained Pendulum branch (Table 3). One-step NRMSE averages 0.56, i.e. the model explains a substantial fraction of successor variance. By the usual reporting conventions this would count as a working world model.

Table 2: Seed-paired effect of CausalNav relative to its routed expert (SOMBRL on CartPole; its own base controller on Pendulum). Positive favors CausalNav.

Cell	$\bar{d}$	95% CI	p	W/10
CP ID	-12.17	$[-101.8, 77.5]$	.766	4
CP shift	11.95	$[-77.0, 100.9]$	.768	5
Pend ID	0.00	$[0.00, 0.00]$	1.00	0
Pend sh.	0.00	$[0.00, 0.00]$	1.00	0

Table 3: World-model diagnostics for CausalNav, mean $\pm$ SD over ten seeds. Pendulum abstains, so no transition model is trained.

Diagnostic	CartPole	Pendulum
Struct. precision	0.53 ± 0.11	–
Struct. recall	0.70 ± 0.14	–
Struct. F_1	0.59 ± 0.09	–
Top-parent acc.	0.60 ± 0.18	0.27 ± 0.21
One-step NRMSE	0.56 ± 0.38	–
Seeds with $\rho > 0$	$7/10$	$0/10$

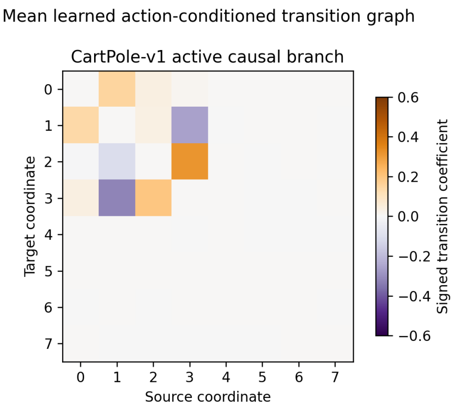

Figure 3: Mean signed pairwise transition coefficients learned by the active CartPole branch over ten held-out seeds. The matrix summarizes *cross-time* coordinate dependencies and is not a contemporaneous DAG. Pendulum is omitted because exact fallback disables transition learning.

It does not translate. Regressing the per-seed paired return difference on per-seed structural F_1 gives $r = -0.15$ ($p = 0.67$) in distribution and $r = -0.18$ ($p = 0.62$) under shift, no relationship, and the point estimate has the wrong sign. One-step NRMSE is positively correlated with the paired difference ($r = +0.71$, $p = 0.02$), meaning the seeds with the *worst* one-step models showed the largest gains. We do not read this as a causal claim; it is a confound, and an instructive one, since the paired difference is driven by the ensemble controller while the fidelity metrics describe a component that is gated out of the argmax. That is precisely the point: two standard world-model quality metrics carried no signal about the quantity a physical-AI practitioner cares about, and one of them pointed backwards.

6.4 Abstention behavior

The certificate is active, not decorative. On CartPole it admits advice on $7/10$ seeds with mean $\rho = 0.32$ and a full $[0, 1]$ range across seeds; on Pendulum it abstains on $10/10$. Abstention there is correct. Pendulum requires energy accumulation and phase-sensitive torque sequences, whereas the library contains three simple temporal patterns per first action. Forcing planning on there was at best neutral and at worst harmful: in development, a low-scale planner reproduced the fallback exactly (-1278.3 ID, -1234.7 shift) while a stronger planner reached -1332.3 ID and -1268.5 shift, i.e. 34–54 return worse (Appendix C). A certificate that converts "my model is inaccurate here" into "do not act on it" recovered the better controller without any test-time label, and it did so from prediction error alone.

6.5 Cost

Planning evaluates $3|\mathcal{A}|H$ model transitions per decision (48 on CartPole, 120 on Pendulum at $H = 8$); transition storage is $O(n^2 + n|\mathcal{A}|)$ pairwise and $O(n^3)$ dense second-order for $n = 8$ coordinates, small relative to the policy network. Mean wall-clock per training seed is 88.3 s for CausalNav on CartPole versus 8.0–40.0 s for the references, and 25.2 s versus 6.4–24.6 s on Pendulum, where abstention removes the planning cost entirely. The CartPole overhead buys the three-member ensemble and the planner; given the paired result, most of it purchased the ensemble.

7 Discussion and Limitations

What we would take to a physical system. The transferable component is not the graph but the gate. A learned model with $F_1 = 0.59$ and NRMSE $= 0.56$ looks deployable by prediction metrics and was nevertheless useless for action selection in one environment and harmful when forced in the other. The certificate detected the second case from prediction error alone, before deployment, with no access to returns or shift labels, and the composition of certificate, margin, and agreement gates makes the failure mode a no-op rather than a regression. For physical AI, we think that ordering matters more than another point of rollout fidelity, and we would want the same ordering in any deployed adaptive controller whose operating conditions drift, from resource-adaptive scaling of running software systems [Ahmad et al., 2025] to environmental response pipelines over nonstationary physical processes [Jois et al., 2026].

Threats to validity. Two low-dimensional classic-control benchmarks, 4,096 interactions per run, and large seed variance mean the returns are not converged and the intervals are wide; ten seeds improve on three but remain thin against bimodal CartPole outcomes. Pendulum uses a disclosed five-torque discretization, so conclusions apply to that version rather than the continuous interface. Shifts change physical parameters but not observation semantics, topology, or reward. Identified coordinates are supervised, so nothing here establishes unsupervised causal representation learning. The references are controlled adaptations, and their names denote inductive biases rather than reproduction-level equivalence. The CartPole branch has more parameters than a single reference expert, so its ranking advantage is partly capacity; a parameter-matched control is the obvious next experiment. The action-count router was chosen after observing exactly two datasets, one on each side of its threshold, and can therefore encode benchmark-specific selection rather than a general law. Acrobot and FrozenLake were examined and not promoted under the fixed budget (Appendix C); this is disclosed because benchmark qualification is itself selection pressure, and a preregistered suite should replace it.

Broader impact. The method targets robust control, where an incorrect causal model can produce unsafe actions. The agreement gate verifies a narrow software non-regression property, not control safety; deployment in physical or safety-critical systems would additionally require calibrated model uncertainty, a conservative fallback controller, and domain-specific validation.

8 Conclusion

CausalNav pairs an identified action-conditioned transition graph with three deployment gates and an action-complexity router, and attains the best average rank (1.25) among ten controlled methods on two benchmarks under physical-parameter shift. The finding we would carry forward is the negative one: structural recovery and one-step error, the two metrics normally used to certify a world model, did not predict whether that world model helped control, while a cheap scale-free reliability certificate correctly withheld it where planning would have hurt. World models for physical AI should be evaluated by their effect on decisions, and should come with an explicit abstention mechanism for the regimes where that effect is negative.

Reproducibility statement

The supplement contains the Gymnasium adapters, all ten agents, the shared trainer, the ten-seed runner, all 200 runs with per-episode returns, the statistics script, and plotting code. The primary command is `python core_codes_v2/run_two_benchmarks.py`; the frozen artifact is `results/public_benchmarks_10seed.json`. Shared modules are deterministically reinitialized from component-specific seed streams so the CausalNav–routed-expert pairing can be audited exactly.

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A Benchmark interfaces and reward accounting

Table 4 records the complete interface used by the runner. No benchmark-specific feature is hidden from a reference method: the padded observation and fixed query are passed to every policy. The causal component uses the first eight padded entries explicitly because the experiment studies control with identified coordinates. PPO receives $0.01r_t$ for numerical conditioning, identically for all methods; every table and figure reports the unscaled Gymnasium reward stored in the adapter's info dictionary. Returns are averaged within 20 evaluation episodes for a trained seed and then across ten seeds; error bars are the sample SD across seed means, not across the 200 pooled episodes.

Table 4: Benchmark interfaces. "Scale" is the divisor applied before clipping each native coordinate to $[-3, 3]$.

Item	CartPole-v1	Pendulum-v1	Shared processing
Native state	$(x, \dot{x}, \theta, \dot{\theta})$	$(\cos\theta, \sin\theta, \dot{\theta})$	Zero-pad to 32 observation entries
Scale	$(2.4, 3.0, 0.20944, 3.0)$	$(1, 1, 8)$	Clip normalized coordinates to $[-3, 3]$
Actions	Left, right	Torques $(-2, -1, 0, 1, 2)$	Integer categorical policy
Query weights	$(.10, .05, 1, .10)$	$(1, .25, .10)$	Pad to eight weights
Query target	$(0, 0, 0, 0)$	$(1, 0, 0)$	Concatenate weights and targets
Episode limit	500	200	Gymnasium default termination/truncation

B Algorithm and hyperparameters

Algorithm 1: Reliability-certified causal navigation

1. Initialize the shared encoder, query encoder, routed expert, value head, and identification probe from component-specific seed streams; initialize the signed transition coefficients separately.
2. Reset the action-sampling random stream after construction so methods with different parameter counts receive paired stochasticity.
3. For each 256-step rollout, store observation, action, reward, done flag, identified state, and the actual identified successor state.
4. Run four PPO epochs, updating the shared policy with PPO, value, entropy, and identification losses. On the active two-action branch, update the transition model with one-step prediction and sparsity losses and clip the two gradient groups separately.
5. Maintain the exponential error of Equation 6; keep $\rho = 0$ during the first 48 transition updates, then freeze ρ.
6. On the active branch, enumerate three length-eight intervention patterns per first action, roll each through the detached transition model, and compute Equation 4.
7. At deterministic evaluation, apply Equation 5 only when the reliability, margin, and agreement checks all admit the advice; otherwise execute the routed expert exactly.

CartPole has no physical no-op, so the neutral action of the intervention library maps to the integer midpoint (the right action). This asymmetry is a limitation of using a common discrete library and is one reason not to interpret the planner as an optimal controller.

C Development evidence and rejected variants

All choices below are excluded from the reported ten-seed analysis. Architecture checks and the exact-copy diagnosis used seeds 0–69; a smoke gate used seeds 70–72; the selected routed variant

Table 5: Frozen hyperparameters shared across both benchmarks.

Quantity	Value	Quantity	Value
Training interactions	4,096	Rollout length	256
PPO epochs/rollout	4	Discount γ	0.99
GAE λ	0.95	PPO clip	0.20
Policy learning rate	3×10^{-4}	Transition learning rate	10^{-2}
Value coefficient	0.50	Entropy coefficient	0.01
Auxiliary coefficient	0.10	Gradient norm	1.0
Slots	8	Slot width	32
Planning horizon	8	Patterns/action	3
Planner-scale init.	1.0	Reliability momentum	0.90
Calibration updates	48	Reliability threshold	0.70
Pairwise sparsity	0.05	Evaluation episodes	20

was then frozen; seeds 73–82 formed an earlier audit; and the reported ten-method comparison uses fresh seeds 83–92.

Table 6: Development decisions. These diagnostics are excluded from the held-out result.

Variant	ID	Shift	Decision
SOMBRL CartPole reference	391.07	394.50	smoke reference
Two-member CartPole route	257.27	247.23	reject
Three-member CartPole route	473.40	462.40	retain
Pendulum base controller	−1278.28	−1234.66	retain exactly
Low-scale causal planner	−1278.28	−1234.66	no gain
Stronger causal planner	−1332.27	−1268.54	reject

Convex hybrids, a two-member route, and planner residuals were rejected. The agreement gate was retained because uncertain or contradictory advice then cannot change the deterministic action. Early qualification also considered Acrobot and FrozenLake: under the fixed 4,096-interaction budget no controlled method learned a useful Acrobot policy, while FrozenLake's discrete grid and sparse reward do not match a continuous identified-coordinate transition model, so reporting either would mostly measure benchmark mismatch.

D Statistical analysis details

The seed is the unit of analysis. For each benchmark and condition let $y_{s,m}$ be the 20-episode mean for seed s and method m; the descriptive SD is $s_m = \sqrt{\frac{1}{9}\sum_{s=1}^{10}(y_{s,m} - \bar{y}_m)^2}$. For the matched comparison we compute $d_s = y_{s,\mathrm{CN}} - y_{s,\mathrm{route}}$ and the two-sided interval $\bar{d} \pm t_{0.975,9} s_d/\sqrt{10}$. Correlations in §6.3 are Pearson over the ten seeds; the Spearman analogues agree in sign and significance (F_1 vs. Δ ID: $\rho_S = -0.23$, $p = 0.53$; NRMSE vs. Δ ID: $\rho_S = +0.59$, $p = 0.07$). Four cells and several method comparisons make uncorrected significance hunting misleading, so we emphasize effect sizes, intervals, per-seed differences, and average rank. No bootstrap over evaluation episodes is used, because episodes within a trained seed do not replace independent training replicates.

E Per-seed results

Tables 7–10 report every seed–method mean behind Table 1, exposing the pronounced bimodality of CartPole and showing that no failed training run was removed. Column abbreviations: CN (CausalNav), TF (TF-Policy), GRU (GRU-World), Split (Split-Latent), GNN (GNN-RAG), EZ2 (EfficientZero V2), SOM (SOMBRL), Flat (FlatMBRL).

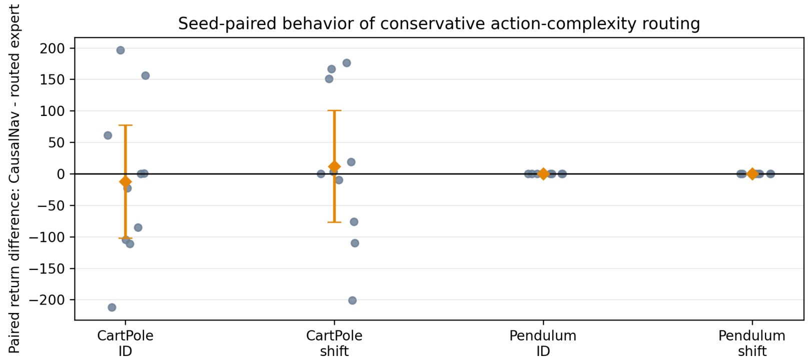

Figure 4: Per-seed return difference between CausalNav and its routed expert (SOMBRL for CartPole; its own base controller for Pendulum). Orange diamonds are paired means; bars are Student-t 95% confidence intervals.

Table 7: CartPole-v1 in-distribution return by held-out seed.

Seed	CN	TF	GRU	Split	GNN	ToG	CAASL	EZ2	SOM	Flat
83	496.9	20.3	146.2	35.5	476.8	159.9	336.8	187.3	496.6	421.0
84	359.9	414.1	228.7	238.9	500.0	337.3	250.6	137.9	464.6	140.2
85	500.0	384.6	477.8	175.1	499.5	500.0	450.6	181.7	343.7	486.6
86	500.0	140.2	500.0	481.1	207.6	500.0	287.2	274.5	438.8	199.8
87	388.8	468.8	184.6	495.6	449.6	259.8	185.8	243.3	500.0	210.1
88	300.1	494.9	203.4	426.0	130.9	144.3	172.2	270.9	103.6	449.9
89	169.2	351.1	381.2	500.0	144.2	500.0	101.7	254.2	254.7	205.4
90	288.1	11.4	326.6	183.1	144.2	182.3	349.4	264.6	500.0	201.9
91	500.0	351.1	281.4	248.2	227.9	405.5	500.0	500.0	500.0	150.4
92	219.6	9.8	170.8	37.0	119.8	85.0	239.9	328.0	242.1	263.1

F Per-seed world-model diagnostics

Table 11 gives the per-seed quantities behind §6.3: structural F_1 against the CartPole ground-truth adjacency, the exponential one-step NRMSE, the frozen certificate ρ, and the paired return differences against SOMBRL. Seeds with $\rho = 0$ executed the routed expert exactly.

G Artifact map and implementation checks

The released runner enforces the following invariants: (i) all methods receive identical benchmark seeds, interaction budgets, PPO hyperparameters, and identified-state targets; (ii) the active CartPole route uses three deterministically seeded optimistic experts while the Pendulum route exactly disables the causal component and executes the base controller unmodified; (iii) optimizer state persists across rollouts; (iv) auxiliary gradients reach the observation encoder; (v) transition targets are actual successor states and never connect the final state of one episode to the first state of another; and (vi) evaluation actions come from learned logits without scripted outcome injection.

Table 8: CartPole-v1 physical-shift return by held-out seed.

Seed	CN	TF	GRU	Split	GNN	ToG	CAASL	EZ2	SOM	Flat
83	500.0	24.8	150.8	45.1	446.4	158.6	278.6	179.7	481.4	400.1
84	314.6	343.4	242.8	211.2	500.0	326.5	231.8	167.1	311.2	195.2
85	500.0	388.9	478.7	178.1	495.0	500.0	443.9	175.2	333.4	500.0
86	500.0	142.5	500.0	480.9	208.6	500.0	289.9	266.6	323.4	167.7
87	390.1	436.9	194.7	488.9	447.1	260.8	186.4	242.2	500.0	212.2
88	259.0	492.4	207.8	396.1	121.2	150.7	176.0	272.2	108.2	460.6
89	176.2	330.2	354.4	500.0	148.8	500.0	105.8	248.7	252.1	209.2
90	298.8	17.4	308.7	187.9	147.3	181.4	350.8	259.6	500.0	199.5
91	500.0	343.6	268.4	251.4	220.2	363.9	500.0	500.0	500.0	152.6
92	226.3	12.2	189.8	53.7	112.2	88.9	237.8	330.6	235.8	288.7

Table 9: Pendulum-v1 in-distribution return by held-out seed.

Seed	CN	TF	GRU	Split	GNN	ToG	CAASL	EZ2	SOM	Flat
83	-1259.5	-1285.0	-1463.1	-1259.5	-1245.6	-1361.8	-1414.5	-1144.2	-1316.1	-1392.9
84	-1303.9	-1124.9	-1100.5	-1114.3	-1172.8	-1285.2	-1145.0	-1410.1	-1391.2	-1269.0
85	-1406.0	-1441.3	-1382.3	-1551.0	-1372.4	-1381.5	-1395.8	-1296.3	-1272.0	-1215.2
86	-862.4	-1183.2	-1420.7	-1236.9	-1364.1	-1405.9	-1566.5	-1089.9	-1130.2	-1408.5
87	-1045.4	-1289.7	-1500.2	-1476.8	-1350.1	-1375.1	-1272.5	-1244.2	-1495.2	-1389.8
88	-1358.5	-1312.1	-1231.6	-1315.5	-1304.4	-1214.3	-1146.2	-1151.0	-1239.8	-1348.8
89	-1101.3	-1374.6	-1376.9	-1357.6	-1167.2	-1275.4	-1294.7	-1403.2	-1341.9	-1263.5
90	-1488.4	-1403.9	-1501.6	-1503.8	-1320.4	-1440.0	-1574.9	-1326.0	-1264.0	-1117.4
91	-1062.4	-1336.5	-1238.2	-1274.3	-1336.2	-1139.7	-1325.8	-1357.8	-1331.5	-1272.2
92	-1136.7	-1298.5	-1247.9	-1188.5	-1118.2	-1355.2	-1402.6	-1309.1	-1087.3	-1344.0

Table 10: Pendulum-v1 physical-shift return by held-out seed.

Seed	CN	TF	GRU	Split	GNN	ToG	CAASL	EZ2	SOM	Flat
83	-1331.2	-1312.3	-1329.2	-1343.3	-1323.8	-1387.2	-1454.6	-1262.4	-1361.6	-1410.8
84	-1159.3	-1073.8	-1101.9	-1200.2	-1193.7	-1213.7	-1227.7	-1192.4	-1166.6	-1255.9
85	-1363.3	-1346.4	-1330.2	-1472.0	-1336.7	-1327.2	-1294.3	-1265.0	-1257.2	-1244.7
86	-1054.2	-1160.8	-1362.6	-1158.7	-1336.9	-1437.7	-1497.3	-1098.0	-1162.2	-1382.7
87	-1139.6	-1236.9	-1391.6	-1342.5	-1303.3	-1318.3	-1220.3	-1161.0	-1302.3	-1331.8
88	-1228.2	-1289.2	-1198.2	-1289.5	-1251.2	-1210.6	-1104.0	-1107.4	-1159.5	-1279.3
89	-1157.1	-1289.7	-1295.6	-1313.9	-1233.6	-1187.5	-1293.0	-1347.7	-1213.1	-1271.9
90	-1395.7	-1341.7	-1272.7	-1300.7	-1280.4	-1312.4	-1306.9	-1282.0	-1186.6	-1174.6
91	-1195.0	-1302.7	-1129.4	-1230.9	-1322.2	-1237.3	-1299.7	-1304.7	-1230.6	-1196.7
92	-1163.8	-1230.3	-1115.2	-1163.2	-1096.9	-1244.5	-1317.5	-1256.8	-1201.5	-1312.6

Table 11: Per-seed CausalNav diagnostics on CartPole-v1.

Seed	Structural F_1	One-step NRMSE	ρ	Δ ID	Δ shift
83	0.500	0.426	0.791	0.4	18.6
84	0.667	0.257	0.720	-104.8	3.4
85	0.714	1.427	0.000	156.3	166.6
86	0.500	0.532	0.000	61.2	176.6
87	0.545	0.678	0.031	-111.2	-109.9
88	0.571	0.860	0.232	196.5	150.8
89	0.545	0.149	1.000	-85.5	-75.8
90	0.727	0.282	0.136	-211.9	-201.2
91	0.667	0.323	0.339	0.0	0.0
92	0.500	0.655	0.000	-22.5	-9.5

Table 12: Primary reproducibility artifacts.

Artifact	Purpose
`agents/causalnav_v2.py`	Transition model, certificate, intervention planner, causal agent
`agents/baselines_v2.py`	Nine controlled reference architectures
`envs/*_benchmark.py`	Public benchmark adapters and structural metadata
`trainer.py`	Shared PPO loop, paired initialization, successor targets, clipping
`run_two_benchmarks.py`	Ten-seed training/evaluation driver with resumable JSON output
`assemble_public_results.py`	Cardinality checks, summaries, paired intervals, frozen artifact
`plot_results.py`	All manuscript figures derived from the frozen artifact